

Life in the Trading Zone: Structuring Coordination Across Boundaries in Postbureaucratic Organizations

Katherine C. Kellogg

Organization Studies, MIT Sloan School of Management, E52-544, 50 Memorial Drive,
Cambridge, Massachusetts 02142, kkellogg@mit.edu

Wanda J. Orlikowski

Information Technology and Organization Studies, MIT Sloan School of Management, E53-325, 50 Memorial Drive,
Cambridge, Massachusetts 02142, wanda@mit.edu

JoAnne Yates

Communications, Information, and Organization Studies, MIT Sloan School of Management,
Cambridge, Massachusetts 02142, jyates@mit.edu

In our study of an interactive marketing organization, we examine how members of different communities perform boundary-spanning coordination work in conditions of high speed, uncertainty, and rapid change. We find that members engage in a number of cross-boundary coordination practices that make their work visible and legible to each other, and that enable ongoing revision and alignment. Drawing on the notion of a “trading zone,” we suggest that by engaging in these practices, members enact a coordination structure that affords cross-boundary coordination while facilitating adaptability, speed, and learning. We also find that these coordination practices do not eliminate jurisdictional conflicts, and often generate problematic consequences such as the privileging of speed over quality, suppression of difference, loss of comprehension, misinterpretation and ambiguity, rework, and temporal pressure. After discussing our empirical findings, we explore their implications for organizations attempting to operate in the uncertain and rapidly changing contexts of postbureaucratic work.

Key words: knowledge; information technology; new organizational forms; work practices; trading zone; cross-boundary coordination

Contemporary scholars have suggested that firms are shifting away from traditional modes of organizing to meet new demands for flexibility, speed, and uncertainty. Child and McGrath (2001, pp. 1139–1140) argue that in our new “postmodern world,” economies are based on “flows of information” rather than on materials. As a result, systems are interdependent across firm boundaries, performance is disembodied from ownership of assets, production and communication change rapidly, and new power asymmetries arise as control of tangible assets loses influence to control of information. In the context of such economic and organizational shifts, the core competencies of firms become dependent on adaptive capacity rather than specialized routines (Rindova and Kotha 2001, Stark 2001, Volberda 1996), and on horizontal collaborations of diverse groups rather than vertical chains of command (Barley 1996).

This changing context raises an important question for organization theorists—do participants in emerging organizational circumstances use the same processes of coordination and knowledge sharing that are effective in more stable and hierarchical settings? And, if not, how do organizational actors coordinate across boundaries in the postbureaucratic conditions where operations are emergent and fast changing, goods and services are

intangible and informational, authority is distributed, and accountability uncertain?

Prior organizational research has yielded valuable studies of cross-boundary coordination and knowledge sharing and the challenges that arise from differences in meanings, norms, and interests. Various approaches have been proposed for dealing with these differences, including transfer mechanisms that rely on a common lexicon and standard protocols, forms of translation that create shared understandings, and transformation processes that generate integrative knowledge and boundary objects (Adler 1995, Bechky 2003b, Carlile 2002, Dougherty 1992, Hansen 1999, Pawlowski and Robey 2004). This research has yielded important insights into cross-boundary interactions and the “artful integrations” (Suchman 1994) often required to accomplish coordinated work across structural, cultural, and political boundaries.

While useful, this research has largely been conducted in traditional hierarchical organizations, such as are found in manufacturing and professional services industries. Some scholars suggest that these traditional bureaucratic structures—with their routinized repertoires—are increasingly less able to respond effectively to new conditions of volatility and virtuality

(Child and McGrath 2001, Ciborra 1996, Heckscher and Donnellon 1994, Neff and Stark 2003), where boundaries and meanings are fluid, emergent, and ambiguous—a set of conditions that Callon (1998) refers to as “hot situations.” In response, firms are experimenting with dynamic strategies of “continuous morphing” (Rindova and Kotha 2001) and “permanently beta” (Neff and Stark 2003), and forms of organizing that are more open, interconnected, and adaptive (Ciborra 1996, Wittel et al. 2002).

One such postbureaucratic organizational form is the “heterarchy” (Powell 1996, Stark 1999), a form comprised of “flat organizational structures with distributed accountability, decentralized decision making, and multiple (often competing) evaluative principles” (Neff and Stark 2003, p. 175). Girard and Stark (2002) note that heterarchies involve relations of interdependence with considerable heterogeneity and limited hierarchy, where workers with different functional and occupational affiliations work on short-term projects that represent the “collaborative efforts of diversely skilled people” (Grabher 2001, p. 361). Such organizational forms rely more on horizontal than vertical relations, utilize multiple, shifting centers of expertise and accountability, and work through rapid assignments, temporary agreements, and creative misunderstandings rather than standard routines and stable commitments (Girard and Stark 2002). As evident in Table 1, heterarchic organizations are markedly different from traditional hierarchic ones.

New organization forms, such as those represented by heterarchies, are becoming more prevalent as Internet startups, biotech ventures, and new media companies dot the corporate landscape (Powell 1996, 2001; Girard and Stark 2002; Heydebrand and Miron 2002; Wittel

et al. 2002). Organizational knowledge about how work is accomplished in these different kinds of contexts, however, remains relatively limited. More specifically, little research has focused on how knowledgeable work is coordinated across boundaries in such new organizational conditions. This is the question we explored in the study described below, which examined cross-boundary coordination in an interactive marketing firm facing the issues of velocity, interdependence, disembodiment, and distributed authority characteristic of heterarchic organizations.

Our investigation leads us to suggest that cross-boundary coordination in heterarchic conditions is constituted by practices of display, representation, and assembly across community boundaries, which both accommodate and resist the dynamics and interdependencies entailed by contemporary modes of organizing. After developing these grounded findings, we join them with recent insights from the social studies of science on “trading zones” (Galison 1997) to propose a perspective for understanding boundary-spanning coordination in the fast-paced and volatile working conditions of post-bureaucratic organizations.

Perspectives on Coordination Across Boundaries

The organizational literature on coordination and knowledge sharing across boundaries can be understood in terms of three primary perspectives (Brown and Duguid 1998, Carlile 2004, Orlikowski 2002, Spender 1996, Tsoukas 1996). The first perspective is informed by an information-processing orientation (Galbraith 1973, Lawrence and Lorsch 1969), where community members are seen to be instrumental in their knowledge-sharing behaviors, acting on the basis of strategic choice and cost-benefit calculations, albeit within the limits of their bounded rationality (Simon 1982). Knowledge is posited as “objective,” that is, external, explicit, and capable of being codified, captured, stored, retrieved, and transferred across people and contexts. Difficulties in coordination across boundaries arise as a result of breakdowns in knowledge transfer due to incompatible codes, routines, or protocols—what Carlile (2004) refers to as a problem of syntax.

Knowledge sharing in these circumstances is enabled by the creation and use of a common lexicon (Carlile 2004) or standard operating procedures that prescribe a means for sharing information (Grant 1996, Nelson and Winter 1982). Individual gatekeepers can integrate differential knowledge by serving as information channels among communities (Allen 1971), while knowledge sharing across teams can be facilitated through the active involvement of multiple specialties in all phases of development (Lawrence and Lorsch 1969). In uncertain environments, specialists

Table 1 Distinguishing Features of Hierarchical and Heterarchical Organizations

Organizational dimension	Hierarchical organization	Heterarchic organization
Form of organizing	Permanent hierarchy	Temporary work teams
Decision authority	Centralized	Decentralized
Accountability	Fixed, top-down	Shifting, distributed
Relations	Vertical, dependent	Horizontal, interdependent
Division of labor and roles	Stable, specialized	Dynamic, blurred
Boundaries	Clearly specified, persistent	Fuzzy and permeable
Work process	Routinized, standardized, rule-based	Improvised, flexible, participative
Composition	Homogeneous	Heterogeneous
Performance criteria	Established, singular	Emergent, multiple
Watchword	Stability, inertia	Speed, adaptability

may develop information artifacts—repositories, specifications, standards—that support communication across boundaries (Adler 1995, Hansen 1999, Leonard-Barton 1992, Nonaka 1994, Teece 1998).

The second perspective highlights the cultural aspects of boundary spanning. Scholars writing about communities of practice (Brown and Duguid 1991, Lave and Wenger 1991, Wenger 1998) suggest that the way work is done within communities reflects occupational conventions and understandings rather than rational calculations of efficiency. Community members use their expertise within particular contexts of action (Lave 1988, Schon 1983), and knowledge is embedded within members' skilled performances and shaped by the community's values and norms (Bechky 2003a, Brown and Duguid 1991, Dougherty 1992, Gherardi and Nicolini 2000, Lave and Wenger 1991). Such knowledge is seen to be largely tacit, situated, and experiential, and not easily articulated or codified. Coordination difficulties arise because of differences in meanings, assumptions, and contexts—a problem characterized by Carlile (2004) as one of semantics.

Knowledge sharing in these circumstances may be aided by the use of a shared language (Bechky 2003b, Lant 1999, Sitkin and Brown 1999), collective stories (Boland and Tenkasi 1995, Brown et al. 1991, Orr 1996, Wenger 1998), or common artifacts (Levina 2001). These both strengthen a community's own perspective and enable community members to acknowledge and appreciate the perspectives of other communities. Differences may be negotiated through the activity of translation, including the use of boundary objects (Bechky 2003a, Henderson 1991, Star and Griesemer 1989) or the assistance of boundary spanners (Brown et al. 1991), translators (Yanow 2000), brokers (Hargadon and Sutton 1997, Pawlowski and Robey 2004), or mediators (Orlikowski et al. 1995).

The third perspective emphasizes the political aspects of knowledge and focuses on what is at stake for communities when members engage in cross-boundary coordination—what Carlile (2004) refers to as a problem of pragmatics. In this view, knowledge is rooted in the accumulated experience and know-how of members, and invested in communities' ways of doing things and measures of worth (Carlile 2004). Because knowledge is seen to be inseparable from people's interests and actions in specific contexts, knowledge sharing requires individuals to modify some of their existing knowledge as they engage in a process of knowledge transformation. Because such a process is contested, it requires investments in time and relationship building, as well as compromises in valued community practices, interests, and jurisdictions (Bechky 2003a, Carlile 2002, Levina 2001).

Differences in interests may be negotiated through the use of shared communicative genres that enable participants to use their relative positions on cross-functional

teams to challenge and transform knowledge in collective action (Levina 2001). The use of common models, maps, and devices that demonstrate systemic approaches to cross-functional issues may help to address problematic boundary conditions (Bechky 2003a, Carlile 2002, Star and Griesemer 1989). As Carlile (2002) argues, effective boundary objects need to provide practical, political means for bridging boundaries: establishing a shared syntax for representing differences and dependencies at the boundary; furnishing ways for individuals to learn about each others' differences and dependencies; and facilitating a process of transforming localized knowledge into novel jointly produced knowledge that transcends each community's local interests, creating in effect a form of "common knowledge" (Carlile 2004), or what Bechky (2003a) calls "common ground."

Looking across these three perspectives, we see that they propose facilitating cross-boundary coordination through the construction of shared commitments (common ground or common knowledge) and the use of various boundary-spanning mechanisms (e.g., routines, languages, stories, repositories, and models). Developing these commitments and mechanisms requires forging agreement around standard procedures, shared protocols, or boundary objects; and such considerable investments in form may be less effective in nonhierarchical and shifting contexts, where criteria of worth are contested, and where areas of jurisdiction are blurred. Indeed, Bechky (2003b) and Carlile (2004) suggest that existing understandings of boundary spanning may not apply in dynamic conditions, to nonmanufacturing firms, or within loosely coupled and heterogeneous communities. Consequently, we do not know how cross-boundary coordination is accomplished in diverse, fast-paced, and volatile workplaces that characterize postbureaucratic organizational forms. This is an important empirical question that motivated our study.

Research Site and Methods

Our research study was inductive and focused on generating theoretical insights from an in-depth examination of cross-boundary coordination in practice. Following an emergent strategy, we collected data from multiple sources in one research site, and then analyzed them qualitatively through an iterative process of data examination, grounded coding, and comparison with existing frameworks. By integrating grounded theory with insights from the literature, we develop a more general understanding of the nature, form, and consequences of cross-boundary coordination in postbureaucratic workplaces.

Research Site

Adweb¹ belonged to the new breed of firms emerging out of the Internet boom of the late 1990s. Formed

in 1995, it crafted custom-built, Web-based, interactive marketing solutions for its clients, including Web-based “brochureware” or “store fronts” for businesses trying to establish a Web presence, as well as end-to-end, enterprisewide electronic business applications that require considerable transaction-processing capability and integration with clients’ existing systems. In 2000, Adweb employed 650 people worldwide, and had offices across the United States and Europe. In the same year, it generated revenues of about \$200 million, having experienced an average annual revenue growth rate of 175% revenue since 1998.²

Adweb competed in a highly uncertain and rapidly changing environment. Along with its competitors, Adweb faced new challenges associated with developing products in the unprecedented and volatile virtual marketplace represented by the Web. The previous five years had seen significant changes in the nature of services, expertise, technologies, and level of complexity demanded by clients. Whereas corporations had initially hired consultants to provide either website design or internal computer systems development, by 2000, they were demanding integrated services to help them deal with the advent of electronic commerce. Such changes increased the technical complexity of projects, required the cooperation of members with different sets of expertise, and obliged firms such as Adweb to stay on top of constantly changing information technologies.

As a business, Adweb operated squarely in the new media sector (Grabher 2002), and its mode of organizing included many elements of heterarchic organizing (Girard and Stark 2002, Heydebrand and Miron 2002, Wittel et al. 2002). All Adweb teams used information technologies extensively, both to create clients’ websites

and to coordinate with one another during projects. Work was performed in temporary, self-organizing project teams operating with fluid authority relations. The teams were composed of members drawn from each of four specialty communities—Project Management, Client Services, Creative, and Technology—the boundaries of which were clearly defined. Members of the four communities performed distinct activities, engaged unique areas of expertise, enacted distinctive identities, were invested in particular sets of interests, and used multiple artifacts (see Table 2).

Members of Adweb’s four communities drew on different areas of jurisdiction and specialized knowledge as they engaged in their everyday activities. Client Services members had authority over identifying client requirements over time, developing client presentations and documentation, and maintaining client relationships. Project Management members created project work plans, developing milestones and measures, monitoring progress, and adjusting plans and schedules over time. Creative members were responsible for the overall aesthetics of the website. They designed visual images, generated graphic concepts and vocabularies, and created user navigations and Web process flows. Finally, the Technology members wrote, reused, revised, tested, and debugged program code, and interfaced with technology standards, technology vendors, and client IT groups.

Members of each community also had different identities, which they expressed through their distinctive work styles, languages, temporal rhythms, and social relations. Project Management members tended to adopt the interests of Adweb as a whole. They pursued organizational values of “on time,” “on budget,” and “to spec.” Members of the Client Services community aligned with

Table 2 Community Groups at Adweb

	Client Services	Project Management	Creative	Technology
Identity	Guardian	Planner	Designer	Builder
Expertise	Building relationships	Managing projects	Designing websites	Building websites
Interests	Maintaining smooth relations with clients and increasing client revenues	Delivering products on time, on budget, and to specifications	Generating innovative, aesthetic, and breakthrough designs	Building robust, reliable, and scaleable websites
Activities	Identifying and assessing client requirements over time	Creating project work plans	Brainstorming ideas	Reusing, writing, revising, testing, and debugging program code
	Developing client presentations and documentation	Developing milestones and measures, and monitoring them over time	Designing visual images	Interfacing with technology standards, technology vendors, and client IT groups
	Maintaining client relationships	Adjusting plans and schedules over time	Generating graphic concepts and vocabulary	
Artifacts (italics indicates community-specific tools)	Word, Excel, PowerPoint, E-mail (Outlook), Calendar (Outlook), Project Management (Project), Intranet, Extranets, Internet	Word, Excel, PowerPoint, E-mail (Outlook), Calendar (Outlook), Project Management (Project), Intranet, Extranets, Internet	Word, Excel, PowerPoint, E-mail (Outlook), Calendar (Outlook), Project Management (Project), Intranet, Extranets, Internet	Word, Excel, E-mail (Outlook), Calendar (Outlook), Project Management (Project), Intranet, Extranets, Internet
			<i>Photoshop, Flash</i>	<i>Java, Perl, SourceSafe, WebSphere</i>

clients, serving as the guardian of client relationships within Adweb. Members of both communities worked long hours, coinciding with those of the team and the client, respectively, and exhibited a work style that communicated efficiency, productivity, and a concern for quality. Both sets of members dressed formally in professional suits, and their language reflected business terminology such as “ROI” and “Target Market.”

In contrast, members of the Technology and Creative communities drew distinctive values, norms, and interests from their occupational communities rather than from Adweb or the client. Creative members were graphic designers, responsible for visual aesthetics, the look and feel and navigational experience of the website. Their goal was to generate a “breakthrough creative vision” that would build their external reputation and win design awards. They saw themselves as self-directed, preferring to work on their ideas independently or through brainstorming with other Creative members. In contrast to team members from other communities, who were colocated by project assignment, Creative members sat in physical proximity to each other, a configuration they guarded jealously. As one member explained, “Sitting together allows for more nurturing [of ideas].” Creative members worked more regular hours than other project members, tending to leave by 5:30 P.M. and rarely working from home, unlike members of the other communities. The language of Creatives included references to “wrapping big thoughts” around websites, or “bringing sites to life” with “sex and sizzle.” They dressed in stylish and trendy clothes, alternating between bright colors in bold patterns, and a monochrome palette of blacks and greys.

Technology members saw themselves as builders, focused on producing robust, scaleable, bug-free websites with appropriate technology. They prided themselves on their technical prowess and aspired to be “alphageeks” (most skillful person in a particular technical area). A member noted: “A true Tech eats, breathes, and lives technology.” The work hours of Technology members varied by project schedules, but often extended long into the evening. While Technology members sat with their project members in a large open area, they donned headphones (with no music playing) to get some private time. They preferred to communicate via e-mail rather than face to face, their dress consisted of jeans and tee shirts or flannel shirts, and they sported cell phones or pagers on their belts. They socialized with each other, both in the office (often playing pool or foosball together) and outside (frequently going for beer after hours).

As Adweb members engaged in their everyday work activities, they enacted community-specific knowledge, meanings, and norms that were valuable for the development of interactive websites. They also enacted community boundaries that posed challenges to project

coordination. A senior manager observed:

It takes a lot of different skills to build an interactive application. ... So, one of the beauties of this industry ... is that you're dealing with these different groups of people with different skills, backgrounds, and experience, education, career expectations, expectations about what their work day will be like, expectations about types of interactions that managers and others will have with them, just the whole nine yards, based on where they've been educated or where they've worked before. ... So, that is a very interesting challenge.

Working across community boundaries was complicated because the activities and outcomes of one community were interdependent with those of other communities. Furthermore, as conditions on projects changed, new sources of difference and interdependence emerged that had to be addressed. Thus, within-community identities, expertise, and interests were “at stake” for community members because they were often reluctant to compromise on investments they had made to ensure their distinctive competence, status, and position within Adweb and their broader occupational communities beyond the firm. Nevertheless, they had to coordinate across community boundaries to achieve the innovative products demanded by their clients, and to do so in compressed time frames.

Research Methods

Our field study focused on the everyday project-based practices of members in one of Adweb's larger offices (110 employees), and was conducted over five months in 2001. Our data collection was exploratory in nature and included interviewing, observation, and review of documents and websites. We interviewed multiple participants within Adweb: senior executives responsible for overall direction and processes, and members of each of the four communities—Project Management, Client Services, Creative, and Technology. We conducted 100 formal and informal interviews (see Table 3 for details). Formal interviews lasted from 45 to 90 minutes in length, and were conducted individually with participants in private offices or meeting rooms. During these interviews, participants were asked to describe their everyday activities, focusing specifically on their work, communication, and temporal practices. We also asked them about their engagement in various projects and key challenges within the organization. All the formal

Table 3 Number and Type of Interviews Conducted Within Adweb

	Client Services	Project Management	Creative	Technology	Total
Formal	10	10	13	12	45
Informal	17	9	20	9	55
Total	27	19	33	21	100

interviews were taped and transcribed. The informal interviews were conducted as participants worked at their desks, or at breaks, and typically focused on specific activities or issues at hand.

Observation involved sitting in on project meetings, shadowing individuals, and hanging out in individuals' cubicles and in the hallways. We followed two projects during the fieldwork, observing each project for its full duration: a three-week business development project that created a "pitch" for Furniture Inc., and a six-week "discovery and planning" project for Insurance Inc. The Furniture Inc. project was formed in response to the client's request for proposals (RFP). Furniture Inc. was an innovative firm with a strong offline brand. As it could not afford to move its whole store online, it was looking for a Web solution to shorten the sales cycle. Eight Adweb members participated in this project: one Project Management, three Client Services, one Technology, and three Creative members. Adweb's winning pitch was for a \$500K project to build an informational website for directing people to Furniture Inc. stores.

On the second project, Adweb was in the initial stages of building an interactive website for Insurance Inc. to include basic information about life insurance and enable information sharing with insurance agents who sell for Insurance Inc. Before the Adweb project, agents used phone, fax, and express mail to communicate with the underwriters at Insurance Inc. The website being developed by Adweb would allow agents to use the Web to log into an Insurance Inc. database, access their accounts directly, and monitor activity on their policies. The project's budget was \$2.2 million, and included nine Adweb members: one Project Management, two Client Services, three Technology, and three Creative members.

In addition to interviews and observation, we collected data by reviewing some of the extensive documentation generated by the organization and team members, including descriptions of the work tasks of the four communities, project plans, schedules, and meeting minutes. We also had access to Adweb's Intranet (internal website), where procedural, organizational, and technical information was posted. These materials provided additional contextual information on project structures, hiring policies, project planning methodologies, and technical specifications.

Our orientation to data analysis was inductive, intended to generate insights into how Adweb accomplished its projects in fast-paced, uncertain, and dynamic conditions. The unit of analysis was practice, defined here as recurrent, materially mediated, and situated social activities (Schatzki et al. 2001) engaged in by members of Adweb communities to coordinate with each other and their clients. These practices were performed by Adweb members as part of the ongoing structuring processes through which they produced and reproduced their community boundaries over time.

This focus on everyday practices allowed us to examine in detail the microlevel interactions that constituted the cross-boundary project work of Adweb members, and how they used technologies to accomplish this. We thus follow Barley and Kunda's (2001) call for detailed studies of what people do in order to "articulate relations between work practices, situational contingencies, and organizational patterns" (2001, p. 85). Our process of data collection and analysis proceeded iteratively, with the early stages being more open-ended than the later ones. We used inductive, qualitative techniques to analyze the data (Agar 1980, Eisenhardt 1989, Glaser and Strauss 1967, Strauss and Corbin 1990), informed by our focus on practices, technology use, and consequences, while remaining alert to emerging ideas. Qualitative analysis is an inherently dynamic and ongoing process, and we cycled among multiple readings of the interview transcripts, field notes, and documentation, iterative coding (in the NVivo qualitative software package) of recurring themes, and the building of categories that helped us identify cross-boundary activities and technology uses in the everyday work of Adweb members. What emerged was an understanding of the distinctive context of cross-boundary coordination within Adweb and the specific *coordination practices* that Adweb members used to get their work done in conditions of speed, uncertainty, and volatility.

Cross-Boundary Coordination Within Adweb

Coordination on Adweb's teams was shaped by temporal, competitive, informational, and technological conditions that characterized their project work. First, team members worked under intense temporal pressure because the Internet world in which they operated had shrunk time-to-market expectations (Castells 1996). Furthermore, the presence of many competitors meant that the products being delivered in highly compressed time frames could not compromise on quality. An Adweb executive noted:

You have to figure out a process where everybody knows what everybody else is doing, and yet you're still doing it fast, fast, fast, which is all clients want. Nothing is fast enough, and yet you cannot make a mistake. There is zero margin for error. . . . You can't sacrifice quality. For a system that's running a company's business, you can't say, "Oh, that error message, don't worry about that one, we have to meet a deadline." You can't say that. It just has to work. . . . Now, clients are suing companies like Adweb up one side and down the other. We're no exception. In this economy and in this environment, . . . you cannot let a single "i" go undotted or "t" go uncrossed.

Second, Adweb members were developing a product whose very nature was changing rapidly over the course of projects, as client requirements shifted, learning occurred, and competition altered the playing field.

A Creative member described a recent project where the strategic direction kept being modified:

So, yeah, I mean we would change direction every couple of weeks. It was like—"We're building a portal. No, we're not building a portal. We're going to have a kids' section. No, we're not going to have a kids section"—you know. So, the objectives of the site were constantly getting changed because the competition was just fierce.

Third, Adweb members interacted extensively with clients throughout the duration of their projects, requiring members to access and act on increasing amounts of rapidly changing information generated at different times and places.³ A Client Services member observed:

I think the amount of client interaction has significantly increased as our touch points with our clients have increased significantly. ... In general I think the work on a project, the work on an engagement, has increased because of that, because of the amount of [client] communication, and that automatically leads to extra work being generated.

Fourth, rapidly changing demands on all Adweb's projects obliged the staffing department to juggle team composition regularly, adding people to some teams while removing them from others. Therefore, expertise and experience were shuffled over time, requiring ongoing information exchange among incoming and outgoing team members. Finally, Adweb members were working with technology (the Web) that was continually evolving, where new standards, protocols, coding languages, interfaces, security requirements, and applications were released frequently, generating creative opportunities as well as critical technical and functional challenges. Kotamraju (1999, p. 466) describes website development as a novel, amorphous, and provisional process that is continually challenged by the "breathtaking speed [with which] the technology of the World Wide Web reinvents itself." A senior Adweb executive characterized this way of working as operating in "the wild wild Web."

Taken together, these work conditions required Adweb project members to regularly access and attend to changing information from multiple sources, and to do so rapidly so as to deal with compressed temporal expectations. For example, on the Furniture Inc. pitch project, team members had to contend with frequent shifts in Furniture Inc.'s goals for the proposed website, its available budget, anticipated target customers, projected sales cycles, and back-end technology systems. This constant inflow of new (often ambiguous) information on the three-week project reflected both Adweb's increasing interaction with managers at Furniture Inc. during the pitch process, and changes in these managers' understandings of their own requirements, those of their customers, the possibilities of the Web technology, and their competitors' strategies and offerings. In addition, the

eight-person team had to cope with a personnel change partway through their project when the senior Creative member, whose experience was needed on another project, was replaced with a junior member.

Coping with a constant flow of information was further complicated by interdependencies among the activities of the four communities and the client. For example, on the Insurance Inc. project, the choices made by the Client Services members working with the client on business requirements affected the work of the Creative member responsible for website navigation. His decisions about navigation then influenced the technologists' design of the technical infrastructure and the other Creatives' ideas for branding opportunities. These decisions, in turn, served to shape the Client Services members' decisions about business requirements. Complicating these decisions was the changing nature of client requirements, as Insurance Inc. members used this opportunity to prioritize their business strategies, and as Adweb members gathered new information about the costs and possibilities of various strategies. Indeed, because of changes in client priorities and Adweb's analyses of various solutions, the specified target users of Insurance Inc.'s website changed three times in the course of two weeks, and the ramifications of these changes cascaded rapidly throughout the interdependent work of Adweb's different communities.

We identified three cross-boundary coordination practices that Adweb members recurrently engaged in to facilitate their dynamic, uncertain, and ongoing accommodation to each other and clients, while retaining commitments to their local identities, values, and interests. In particular, we found that Adweb members *displayed* work across boundaries (i.e., they made their work visible and accessible to other communities), they *represented* work across boundaries (i.e., they expressed their work in a form that was legible to other communities), and they *assembled* their separate contributions across boundaries into an emerging collage of diverse elements (i.e., they reused, revised, and aligned their work over time so as to keep it dynamically connected across multiple communities). We found that these three practices of display, representation, and assembly afforded Adweb members a certain flexibility and rapidity in their project work. However, these gains were not achieved without costs, and we also found the coordination practices entailed tensions, slippages, and problematic consequences as Adweb members both complied with and resisted the imperatives to be flexible, innovative, and "fast, fast, fast." Below, we describe the three coordination practices before discussing some of the difficulties associated with their performance.

Display Practices

Given the rapid temporal pace of most projects, Adweb members typically worked on their contributions in

parallel, requiring them to stay informed about what work was being done by other members of the project. To do this, Adweb members made their work visible to each other in a set of activities we label *display practices*. That is, they regularly made their current work available to each other, and made their time commitments and work assignments transparent so as to facilitate ongoing coordination across the communities. Display practices involved using multiple information technologies, most notably the firm's internal and external networks, electronic mail system, online calendaring tool, and project management system. Members engaged in these practices to stay on top of rapidly changing conditions and time frames, and to be informed about progress on the project.

When participating on projects, Adweb members made their work-in-progress visible by posting it to the firm's internal network (an intranet), sending it through the project's team e-mail, and loading it onto the firm's closed external networks (extranets, one per client project). For example, a Project Management member on the Insurance Inc. project explained that she no longer saved any files to her computer's hard drive, just to the intranet. She also noted that in the past, she had carried around a huge black notebook of important documents, but now simply carries her laptop, knowing she can pull up any file she wants off the network. On both the Furniture Inc. and Insurance Inc. projects, team members posted meeting notes, templates, schedules, and outlines of project documents on the Intranet. They also posted project deliverables (e.g., pitch presentation, discovery matrix) on the intranet and extranet so that team and client members could access these materials at any time. The display of information on projects' extranets kept clients up to speed on project progress and changing conditions. A Client Services member explained:

Absolutely every deliverable and everything is on this Extranet. It's a good archive of everything that's gone on in the project.

Use of extranets also extended the reach of interaction with clients across time and space. As another Client Services member observed:

...And almost all of our clients seem to be traveling 80 percent of their time. The only reason we're able to make progress on these in a timely fashion is because of tools like the Extranet. There'll be a person who's located in Geneva...[and he] is constantly traveling across Frankfurt, London, New York. And oftentimes... I don't even have a clue whether he's in Geneva, in Frankfurt, in London or in New York. But he's looking at the Extranet. Clients know that everything is always on the Extranet. So that as soon as they know there is a [teleconference] meeting, they will automatically go to the Extranet, log on, and the most recent version of the document will be up there on it.

All communities within Adweb also made extensive use of their electronic mail system (Microsoft Outlook), and projects established a mailing list to ensure that all team members on a particular project received all relevant project communication. Project members were expected to regularly send updates to this list about their work, and in this way, everyone was "kept in the loop" about what was happening on the project. The following list indicates the subject lines of a few project e-mails sent on the Insurance Inc. project (sender's community is indicated in italics):

Notes from 6-26-01 Tech Arch Meeting attached; please review/edit and approve [*Project Management*]

Proof of concept [*Client Services*]

Discovery Matrix Schedule [*Client Services*]

Source Control [*Technology*]

Notes from Development Environment Call 6-27-01 [*Project Management*]

PLEASE USE THIS TEMPLATE FOR DISCO DOC SECTIONS [*Client Services*]

Insurance Inc. Status Report 7-3-01 [*Project Management*]

Such use of project e-mail helped to keep everyone informed about the different communities' activities and the progress being made on the project as a whole, thereby facilitating coordinated work and ostensibly keeping all project members "in the know." For example, when the Client Services member on the Furniture Inc. project received updated budget information from the client, she sent a project e-mail about it, indicating it was on the low side and describing several other low-budget websites that Adweb had built in the past, and what had been done to keep costs down.

In addition, because project members were available via e-mail at almost all times, matters that required rapid response could be expedited via the project mailing list. One Technology member explained:

Accessibility lets you get things resolved quicker or escalate things and get the right people involved much quicker. ...The pervasiveness of e-mail as a business tool is SNAP. Everything's immediate.

Some members noted that this immediacy and visibility allowed them to keep an eye on things from home, thus allowing for a greater sense of control over their work:

In the past what would happen is that you'd be home and you'd have a lot of anxiety because you'd know that there was shit going on but you didn't know what that shit was, and you weren't going to know until the morning. I know some people view [using e-mail from home] as an infringement on their time. I view it as a sleeping pill. Go on [the network], see what went on, if there are any fires to put out, you have an opportunity to do that, so when you come walking in in the morning there aren't these time bombs that have been sitting there for X number of hours.

Adweb members also displayed information about their temporal commitments to each other, and they did this through their collective use of an online calendaring system (Microsoft's Outlook), stored on the firm's Intranet. This online calendar gave project members access to the schedules of all other team members, allowing them to find out who was doing what and when, and who was available at what times. One Project Management member referred to it as "the team information area," as it provided a collective window onto project events, meetings, and deadlines, indicating times of overlap and conflict. Members used the calendar to book time with people, organize project meetings, and arrange for client teleconferences. In Adweb's relatively flat organizational structure, booking someone's time was an implied request that they attend the meeting; it was not interpreted as a demand. Project members relied heavily on calendaring to coordinate their own and others' activities. One member reported relying on it "to tell me where I have to be, when," while a Technology member explained that it sped up cross-boundary interaction:

It does help expedite a lot of things. You can just quickly look at everyone's calendar, say, here [pointing to screen] and tell if [they] can or can't make it. And we use that heavily. So I think it does speed up things, without a doubt, all the way around.

Project members also made their work assignments and project deliverables available to each other through the use of an online project management tool (Microsoft's Project). This helped them to develop and access highly detailed plans and timelines (stored on the firm's intranet), which they then used to orient their ongoing activities and adjust to shifting interdependencies over time. On the Insurance Inc. project, for example, the team maintained two levels of project plans: long term and short term. The long-term plan included actual and proposed hours for each person on the project, their billable rates, and "high-level" weekly tasks for their group; while the short-term plan included specific tasks (including priority level and expected completion date and time) for each member for each day.

While Project Management members typically created project plans and schedules, these were available to all project members who would access them online to monitor individual responsibilities, community obligations, and overall project status. Ongoing monitoring of their own and others' activities and commitments through these online plans helped members keep projects on time and within budget. It also ensured that no single person was in charge of all the planning and monitoring for every project, allowing some flexibility in the assignment of people's time to projects and enabling a distributed authority in accomplishing the work. Thus, different members took the project lead at different

times, depending on client requirements and the nature and phase of the project. A Creative member noted:

Someone is always playing time cop. But it is not necessarily the same person every time. For the Information Marketplace project it was just a creative gig so Creative was top dog. ... Whereas Financial Services Inc. was a technical project.

The rotation of the project lead was thus facilitated by all project members having online access to all the project plans, schedules, and commitments.

Display practices facilitated the distributed engagement of diverse participants at multiple times and from numerous locations. By constantly making information about work-in-progress, commitments, plans, and schedules available to others, both within and outside of the firm, Adweb members were able to continue working on their specific contributions, while connecting with other project work and attending to emerging issues.

Representation Practices

In addition to making their work visible to each other, Adweb members made their work legible to each other through a set of coordination activities that we label *representation practices*—expressing ideas and concepts in a particular form that can be used by others. This practice differs from translation, which involves constructing shared meanings across communities. Rendering work legible, in contrast, requires articulation in a form that is tangible, observable, and readable by others (Callon 2002, Iskander 2005), even if readings are multiple and differ from those intended by the author. As Wittel et al. (2002, p. 192) found in their study of heterarchic firms, the imperative to move quickly and stay connected to emerging trends results in an emphasis on operationality rather than "meaning and interpretation." Similarly, at Adweb we found project members focused on "keeping the work going" through shared *forms* of communication rather than shared *content*, drawing on a repertoire of cross-boundary project genres to represent their work to each other.

Genres of communications (e.g., memos, letters, meetings, reports, and presentations) follow discursive norms that suggest the use of certain expressive templates in particular recurrent situations (Orlikowski and Yates 1994). Collective use of genres facilitates communication within and across communities (Yates et al. 1999). In Adweb, members enacted and circulated a number of project genres that facilitated their cross-boundary information flow. This repertoire of genres (Orlikowski and Yates 1994) represented both the work being done (e.g., presentations, requirement specifications, prototype websites) and the process of doing so (e.g., status reports, project schedules). Many of these genres were enacted with the various information technologies used by Adweb members (e.g., Microsoft's PowerPoint, Excel, and Word).

A dominant project genre used by Adweb members to represent their work to each other and clients was the PowerPoint presentation genre (Yates and Orlikowski, forthcoming). Within Adweb, this genre was instantiated in a presentation template that incorporated the standardized layout, syntax, and default bullet-point format inscribed within the PowerPoint technology. More specifically, the “pitch presentation” was enacted by Adweb members to propose new work to a prospective client. The template for this genre included PowerPoint slides for the following topics:

Project Overview; Business Overview; Client Need; User Scenarios; Creative Ideas; Technology; Adweb (Past Clients, Prior Projects); ROI (Return on Investment)

Members’ use of the presentation genre normalized the representation of ideas across communities and projects, enabling members to articulate and simplify their contributions in a manner that rendered their ideas and interests legible to each other. An Adweb senior executive noted that using such presentations forced a kind of clarity within the project, particularly in communication with clients:

You can’t hand the senior executive a 200-page document. ...I think [PowerPoint] forces people to try to decide what they think is important and to dare to be wrong. If you’re going to show clients stuff iteratively, which you jolly well better, you have to dare to be wrong.

Representation work was complicated by rapid changes in the external environment and internal team composition that led to frequent changes in community-based expertise and team conceptualizations of the product. For example, partway through the Furniture Inc. pitch project, a new Creative member was assigned to the team. This member knew how to use Flash, Macromedia’s Web technology that allows developers to integrate video, text, audio, and graphics into rich and interactive interfaces. With the new infusion of expertise in Flash technology, the three Creative members began to conceptualize the product differently, and wanted to embed a Flash demonstration into the pitch presentation. Even as these Creatives began work on their Flash segment, a Client Services member received new information about the potential value of online customers to Furniture Inc., and began to work on incorporating a new ROI analysis into the pitch presentation. Because of the rapid changes in understandings of the product across communities, it would have been time consuming and difficult for project members to continually negotiate their differences and reach a common consensus. Instead, Creative members represented the potential of the proposed Flash technology for the Furniture Inc. project by drawing on the pitch presentation genre to communicate a stripped-down version of the Flash segment to the other team members. Similarly, the Client Service members used the pitch presentation genre to

represent their new ROI analysis in a simplified manner, highlighting its general implications for the marketing of the proposed website. By representing versions of their work to each other through their contributions to sections of the pitch presentation genre, members continued working on their separate areas without getting full agreement on the particular details. In this way, elaborated versions of both the Flash segment and the ROI analysis were eventually incorporated into the final pitch presentation that was delivered to the Furniture Inc. clients.

Members also used other project genres to represent their work to each other and the clients. On the Insurance Inc. project, Adweb members worked closely with client members to develop a standardized “proof-of-concept” (POC) of the website. This genre involved building several simple and standardized Web pages that demonstrated Adweb’s capability to connect to clients’ back-end systems. To develop the POC for Insurance Inc., Technology members at Adweb wrote Java code running on IBM’s Websphere (a software platform that supports electronic business applications), and connected it to a set of client-built system components that linked the Adweb POC Web pages to Insurance Inc.’s databases. Use of this POC genre demonstrated technical connectivity without requiring all the specific details to be understood or worked out. Its representation remained simple enough so that non-Technology members from Adweb, as well as the client, could understand and comment on the features of the potential website.

The discursive norms of project genres provided expectations about communicative purpose, participant roles, and domain areas, thus outlining the cross-boundary coordination process to be followed. This was particularly helpful in conditions of rapid change and shortened timeframes. For example, the “discovery matrix” genre, generated in a Microsoft Excel template, contained several categories of website features such as content, look and feel, reporting, printing, searching, etc. Using this matrix (stored on the Extranet), client members set business priorities (low, medium, or high) for every feature to be included in their website, and Adweb members then assigned details and levels of effort to the selected features. On the Insurance Inc. project, members knew and played their part—without explicit discussion—in the collective enactment of the discovery matrix genre. Thus, the information architect (a Creative member) completed the particular rows that detailed the specific reports the system would generate for the users, and once he was done, the Technology members estimated the technical effort required to generate those user reports, color-coding the corresponding cells in the discovery matrix according to effort level. Once completed, the discovery matrix provided a common framework of what needed to be done to develop

the Insurance Inc. website, by whom, when, and in what sequence.

After the discovery matrix, the team generated another project genre known as the discovery document (or “Disco Doc” for short) that detailed the client’s requirements in various functional areas and proposed recommendations for the development phase of the project. This multipart Word document was stored on the firm’s intranet so that members from the different communities could access it at various times to work on their sections. On the Insurance Inc. project, this document was quite extensive (128 pages) and included the following types of content, as listed in the table of contents:

Project Overview; Business Overview; Creative Ideas; Functional Requirements; Technical Summaries; Definition of Deliverables; Timeline; Financial Summary; Future Considerations

Adweb members also used the genre of project meetings to manage the flow of information across their different communities and with clients. For example, on the Furniture Inc. project, the Project Management member scheduled “check-in meetings” three times a day. Typically, the morning meeting focused on user scenarios, the afternoon meeting on design ideas, and the late afternoon meeting on the evolving pitch presentation. On the Insurance Inc. project, the team had weekly status meetings with the client via conference call. Before each call, a Client Services member would write up a status report, post it on the project’s extranet, and send a copy to the team via e-mail. For example, an agenda for one weekly status report conference call included the following details within a prescribed set of topic areas:

Last week’s goals: Revising site map

Summary of progress: Delivered the Creative brief

Goals for this week: Receive input and approval from Insurance Inc. on revised site map by Tuesday, July 10th

Open issues: Further detail of how users interact with screens is required prior to the design phase

Summary of project milestones, dates, and status: [left open]

Notes from the project status meetings (another project genre) were written in Microsoft Word by a Project Management member, posted to the firm’s intranet and client extranet, and sent via the project e-mail list to all Adweb team members.

The use of project genres within Adweb structured members’ cross-boundary communication by providing a repertoire of socially recognized and legible templates that helped the flow of information and ideas across boundaries. Adweb members used these genres both routinely and frequently, enabling them to represent their community-specific ideas and issues to each other without requiring general translation or collective agreement. Reliance on such representations allowed members to coordinate both their distinctive as well as their parallel areas of project work even though local meanings and interests were not shared.

Assembly Practices

Adweb members contributed to their project’s collective effort through a form of working together that we label *assembly practices*. These work practices involved project members referring to, reusing, revising, and aligning the work products of other communities in their construction of interdependent products. In contrast to collaborative efforts that attempt to produce joint products, assembly work involves the juxtaposition of separate elements that, while they are not tightly woven into a single narrative, do express orderings that relate to and reference each other. The result is a dynamically constructed text with interrelated references and contingent links (Borer 2002). Giddens (1991, p. 26) refers to this dynamic juxtaposition of loosely linked items as the “collage effect,” arguing that it signifies a central aspect of mediated experience in contemporary life. At Adweb, we found members recurrently accessing, reusing, and revising each other’s diverse contributions and assembling them into a collage of various products such as client presentations, discovery documents, and websites. Use of Adweb’s online networks facilitated this cross-community work by creating a common, online digital space within which members could locate, learn about, and engage with each other’s work. Use of such a digital space also accelerated work, as members had round-the-clock access to it, allowing dynamic and ongoing assembly of project deliverables.

On the Furniture Inc. project, for example, two members from Client Services and one from Project Management worked concurrently via the firm’s intranet to create the pitch presentation. The presentation (in PowerPoint) was saved on the intranet so that the members could view the latest version at any time, but only one of them could work in the file at a given point in time. The two Client Services members began by dividing up areas of the presentation for each to do. However, as they each worked at different points in the file, they often modified each other’s slides in order to make them consistent with their own work. Also, as they worked on the presentation, they repeatedly e-mailed the Project Management member to ask her for specific information for particular slides. She would then access the file on the intranet, display it on her desktop computer, and enter the requested information directly into the emerging client presentation.

Similarly, team members on the Insurance Inc. project assembled the discovery document by adding their contributions to designated sections. As they did so, their changes often had implications for sections being worked on by members of the other communities. For example, the work of the Creative members designing the website had implications for the technical building work of Technology members, yet these two communities needed to work concurrently to meet their compressed time frames. When members made changes

that they knew affected others, they did one of two things: If they believed their changes were uncontroversial and unlikely to be contested, they made them directly to the affected section and then e-mailed the other members notifying them of the changes; if they were unsure about the implications of their changes, they first e-mailed the other members to discuss the changes. This process usually worked well; as a Project Management member observed, “People are good about checking in when they have made changes.” He also noted that project members used a version numbering system to indicate changes in the collective document.

One implication of collective assembly for everyday work was that members needed to be willing to reexamine and revise their contributions on an ongoing basis. A Client Services member explained:

Information makes things more egalitarian. People have access to contradictory information. It means you need to tolerate a higher level of critique and review. You need to pick your battles. Everything is pretty open and transparent.

Creatives, particularly, deemed this process of ongoing revision as critical to staying innovative in a competitive market. Two Creative members commented:

We’ve only been around [...] since 1995. So no one can say “I have seen it all and this works best.” Some people try to say that the Web has been all figured out. But [we] say, “You are out of your mind.” It has to be fluid because it hasn’t stopped mutating.

No one has figured out the process yet. ...[so] we rapid prototype and iterate.

Prototyping also served to accelerate turnaround time with clients, as a Project Management member noted,

Now we are giving the client a new release every day. So we make improvements to it and give it to them every night so they can see the improvement in it.

Assembly practices included the activity of reuse as well. Before the availability of the firmwide intranet, Adweb members were likely to treat each new piece of work as if it had never been done before, or to reuse only work that they had personally generated in the past and had available on their own computers. With the availability of the online electronic archiving features of an intranet, Adweb members began posting documents associated with various aspects of project work in a firmwide repository, and this displayed work became a valuable source for information and ideas on future projects. For example, a Client Services member described and demonstrated how she reused work from prior projects:

When we do a pitch, we need to show the potential client what kind of work we do. So I need to pull old cases. Usually I start with cases I have worked on and have saved on the network, like this one [points to screen].

Next I go to the Intranet and look at rooms [as project files were called] on a particular topic. Like now I am looking for a case on Customer Relationship Management, so I go to that room and click on “case studies.” I use this room a lot. OK, now I want telecom. I can’t find it in the room; telecom is a new industry for us. I know we also have something on our Intranet called the A-files which lists cases by topic and has specific information by case, ... so I’ll go there.

In generating and reusing such a repository, Adweb members had access to a broad set of prior solutions that they could leverage on their current assignments, facilitating distributed assembly that could engender a faster response on projects. A Project Management member explained:

There’s a lot of things that technology provides that make unnecessary a lot of things that proposal teams like ours would have had to worry about previously. There is a foundation of knowledge there on our Intranet, lessons about how things were done in the past. And on the Web there are tools that describe how processes should be executed. There’s a lot that doesn’t have to be recreated with projects or proposals.

The availability of documents on the firm’s intranet facilitated the assembly of work across time, clients, and functional areas as information was extracted from prior projects (decontextualized) and then included in current project work (recontextualized). As this work was typically represented in a form that had been “road-tested” on earlier projects, it did not require extensive “reprocessing.” A Technology member commented that his reuse activity afforded certain efficiencies:

Since we are building multiple versions, and have an installed version working already, we can pull code from this version to use in our version. We try to reuse code whenever we can, and go back to the same network folder a lot. We didn’t write this code, but it is easy to find the small fragment we are looking for. ...In the past, we would have needed to go to the developer and ask him for his code.

Project members also used extranets to facilitate distributed assembly of project work with clients. A Project Management member commented “There’s definitely more ongoing negotiation with clients now that we are working more closely with them,” and a senior executive explained:

What happens is the dependencies and the interconnections between what you’re doing and what the client does now need to be raised to a level where it’s highly visible and always looked at. ...It’s much more at the surface now than it used to be.

A Creative member offered another illustration:

Like when we present first round Creative...there’s always more people that need to know about it behind the scenes so, what we’ll do is we’ll schedule a conference

call, we'll have the Extranet open, we'll walk them through everything, and then we'll hang up and they can go off and show whoever they need to show. They can reference it themselves later and then they can go right in the comments box and e-mail us if they have concerns or they see something they like or didn't like, that they didn't notice in the meeting.

On the Insurance Inc. project, for example, Adweb members engaged in distributed assembly work with clients in developing interview questions. Team members posted suggested interview questions on the project's extranet, and the project liaison at the client site revised these questions and posted them back onto the extranet for all the Adweb members to see and comment on.

As they assembled work products, Adweb members dynamically aligned their work by allowing for ongoing adaptation in ideas and effort. They did so by engaging in what Girard and Stark (2002) label "provisional settlements," where team members postpone extensive consensus building on an issue, and negotiate a temporary agreement to keep the work going, with the mutual understanding that these areas will be revisited and resolved further down the line. Provisional settlements helped members proceed in the face of uncertainty while also keeping negotiations flexible and ongoing. For example, in a meeting on the Furniture Inc. project, a Client Services member proposed the use of a new technology LifeFX, as he had some interest in turning it into a cross-selling tool for Adweb. However, one of the Creative members did not think the technology looked good visually and so did not want it in the presentation. After a few minutes of back and forth discussion, the Project Management member brokered a temporary truce between the two members by saying "Let's get through these slides and make a list of what we need to do. We can come back to this point." At a later meeting, members returned to the issue and decided to include LifeFX as a talking point only, rather than as a substantive part of the presentation.

A similar accommodation was reached on the Insurance Inc. project. In this case, a Creative member was working on "user scenarios" (hypothetical descriptions of system usage) in order to propose an overall website map and specific architecture. To develop his user scenarios, he was hoping to fly out to visit the client and spend several days conducting interviews with prospective users, but he was depending on the Client Services members who were analyzing target customer segments to identify types of users to interview. Meanwhile, Technology members were depending on the Creative's decisions about information requirements for the website as they needed to know what fields in the client database to link to. Because Client Services could not yet confirm who the potential users were, project members provisionally settled the matter by agreeing that the

Creative member would delay his trip and conduct several interviews via phone, beginning the website map with the knowledge that it would likely be altered pending new data. In turn, the Technology members shaped their programming work based on the initial information generated by the phone interviews, knowing that this would require some modification in the future. This provisional settlement provided the alignment required to continue assembling work products.

Problematic Consequences of Cross-Boundary Coordination Practices

While members within Adweb worked across their community differences on projects, issues of identity, control, and accessibility occasionally created difficulties for project communication and outcomes. These recurrent issues were particularly evident when conflicts arose over the distinctive values, interests, and norms of the four communities.

Issues of Identity. Conflicts over community identities were particularly evident in display practices where members were expected to render their work-in-progress visible to all on the project. Client Services and Project Management members were very supportive of activities and use of technology that facilitated such project transparency because it facilitated their tracking of project deliverables and client communications, while also enhancing their respective identities as planners and guardians. Technology members were also willing to display their work to others because this allowed them to share code with others, and having others reuse their code boosted their technical reputations and helped them gain the valued status of "alphageeks."

This was not the case with Creative members, whose valued identities were grounded in distinctively individual and novel contributions that were seen to be sole-authored "signature work." As one Creative member observed, "It's all about pride of ownership. You want to advance your vision." Creative members built valued identities by first developing their ideas individually and independently, and then sharing these with others only after they were completed. Not surprisingly, Creative members were concerned about premature display of their work, particularly if they thought this would inhibit their ability to develop original work or make subsequent modifications. Both such constraints were seen to potentially damage their occupational reputations as self-directed, innovative designers. On the Furniture Inc. project, for example, the senior Creative member had begun to interview customers to create "customer scenarios" that depict diverse market segments and user profiles, but she did not post this information onto the firm intranet because she wanted to wait until she had finished her data collection and had concluded her assessment of all the customer information.

For other project members, Creatives' refusal or delay in displaying their work-in-progress was frustrating and perplexing, as a Project Management member noted:

There is always this tension. You budget one week for something in Creative and by day 3 you have nothing. You can't make a person come up with a good idea. And the biggest inhibitor of creativity is fear, so you can't stand over them and badger them. The Creative thing is very strange.

In response, Creatives argued that their work was not predictable or certain, as one explained:

There's just a natural thinking process. It's not cooked until it is cooked. ...Creative is 60% in here [points to her head] and 40% on the page. The 60% comes first and people are frustrated that they can't see it. Our world is not as tangible as the tech world. They have hardware and software, and our stuff comes out of nowhere. It comes out of the other side of the brain.

Issues of identity and the specialized language, expertise, and work styles associated with it were also evident in members' use of project genres. In particular, Technology and Creative members sometimes resisted the use of the PowerPoint presentation genre when they felt that condensing and normalizing representations of their work across community boundaries would lead to a loss of meaning or the forfeiture of valued distinctiveness. On the Insurance Inc. project, for example, a Client Services member asked a Technology member to provide some content for a PowerPoint presentation to be sent to the client, noting "Don't do anything fancy, just a few points." The Technology member contested this, observing to us, "...sometimes things get so reduced to bullet points that both Client Services and the client don't understand what we really mean." Creative members similarly resented the expressive limitations they experienced in using this genre. One Creative member complained that its use "increased mediocrity," while another noted that she found it professionally constraining:

I'm not a big fan of PowerPoint. I hate it that we use such a primitive program. It is such a low-end presentation mechanism for a Creative deliverable.

In a meeting we observed on the Furniture Inc. project, a Client Services member (Kyra) had an exchange with a Creative member (Bruce) about what to include in the pitch presentation. Kyra wanted to use the standard pitch presentation genre, while Bruce wanted to include some animation, arguing that this was the only way to convey the graphic design aspects of the potential website—

Kyra: You're gonna hate me for this, but we want them [the client] to take away that we are a business partner and will increase sales and this [the animation] is a distraction.

Bruce: It is visceral. ...We will build more than just brochureware or the typical blather that happens on the web. A great website needs to be idea driven. ...You can't bullet point this. We march to a different drummer.

In a conversation after the meeting, Bruce noted his frustration with standard genres:

It's hard to teach these account people [Client Services]. They think that we'll wow them [the clients] with a PowerPoint preez. It's such a chest-beating thing. But saying how big we are and how global we are isn't working anymore. ...I mean, really, who cares? Now we need more show biz and more dog and pony stuff.

While resistance to rendering work visible and legible to others reveals concerns over valued identities, it also indicates a concern over issues of control.

Issues of Control. In some instances, members resisted cross-boundary coordination practices in order to retain jurisdictional control over their work practices. For example, in a meeting on the Furniture Inc. project, a Client Services member noted that one of the Creative slides in the evolving pitch presentation was "too busy" and suggested that the intended brainstorming effect might be conveyed by displaying "a lot of words at different angles." To this, the Creative member retorted:

Please don't design the slides. Let the Art Director [referring to himself] do that. You've told me that you want something to indicate brainstorming. I know what you want. Now let me do it.

Commenting on this later, another Creative member observed:

I think that not many people outside of Creative can really add to the creative product. Our writing is for creative output; it is a tone poem, mood selling. Just as it would be inappropriate for me to go in and edit the technical section.

Exercising jurisdictional control was an ongoing issue for all members, not just the Creatives. In a meeting on the Insurance Inc. project, we observed a senior Technology member (Jon) trying to persuade other members to reuse existing materials so as to leverage Adweb's knowledge of the client developed as part of a previous engagement with Insurance Inc. The Client Services member (Carol) argued that this reuse would be distracting and time consuming because it would be difficult to find all the relevant information. No resolution was reached and Jon left before the end of the meeting, at which point Carol remarked to the others that Jon tended to make things a "pressure cooker" and that she did not think that "this thing about knowing their [the client's] business is of crucial importance." While the disagreement in this meeting was expressed as differences about the value of leveraging knowledge versus time efficiency, it was also a contest over control.

Jon was trying to shape communication with the client based on his previous experiences with them, while Carol was trying to protect the traditional interests of her community as the guardian of the client relationship.

Issues of control were exacerbated by the fast speed demanded on projects, and the increased accessibility of project members through the various online tools in use—e-mail, intranet, extranets, and calendaring. For example, use of the online calendaring tool made people's schedules completely transparent to everyone on the project. This meant that project members were able to schedule the time of other project members, leaving individuals with less control over their own time. This sometimes created unintended difficulties when people were double or triple booked for a given time slot. Such individuals were then faced with the dilemma of having to decide which of the various conflicting meetings to attend. We observed a number of occasions when these people were "helped" in this decision by being physically escorted to a particular meeting by someone who had a vested interest in their being there. Creatives resisted the calendaring practices most—for example, by not showing up for meetings and leaving the office by 5:30 P.M. every day so that people would not schedule meetings with them after that time.

Exercising control could also create difficulties when members had to reach provisional settlements as a way to cope with project uncertainty. In some cases, such settlements were not negotiated in a timely fashion or were bypassed altogether. A senior executive recounted how, on an earlier Insurance Inc. project, differences in assumptions between the Creative and Technology members had not been recognized or addressed through a provisional settlement, leading to negative consequences:

Insurance Brazil is an excellent example of where we really screwed up and [only] saved ourselves, sort of, just in time. What happened was the Creative team developed a set of assumptions about what they were doing. The Creative team was building schematics for the screens. . . . So they went and did a great job of doing a really, really encyclopedic deck of schematics, which the client signed off on. Meanwhile, the Technical team went off and developed functionality documentation separately. The two groups didn't check in with one another. And to this day, we are still suffering the consequences.

Proceeding without provisional settlements was often exacerbated by the use of project e-mails where members could inform others about decisions without actually engaging them in discussion. A Creative member described this e-mail practice as "such a passing the buck thing: 'What? Didn't you get my e-mail? You had access to the information; it's not my fault if you didn't use it.'" Similarly, a Technology member recalled the difficulties that arose when a provisional settlement had not been negotiated between Technology and Creative

members:

On my last project, Creative scoped it without involving Technology. Instead, they said, "We're done with the idea and now you have until April 1st to deliver it." It was a nightmare to build. And we got absolutely burned by that.

While these incidents suggest that Creative and Technology members often encountered conflicts over how to proceed, their everyday activities and talk reveal a pragmatic acceptance of provisional settlements. For such members, provisional settlements reflected the emergent nature of their work and allowed them to keep moving forward even though explicit and broad-based agreements on key issues had not yet been achieved. In contrast, Client Services and Project Management members expressed concerns about provisional settlements, both in their actions and their talk, experiencing them as escalating the risk and uncertainty of projects and reducing their control. Because provisional settlements meant that important concerns had not been definitively resolved, there was an increased burden on such members to keep things in dynamic alignment, and not allow the sort of disconnect that happened on the Insurance Brazil project to recur. Both Client Services and Project Management members reported feeling a greater need to be continually available to other project members in order to ensure that things were kept on track. For example, the Project Management member on the Insurance Inc. project related that he "logged on to check-in on burning e-mails" every night after his daughter was in bed, so that he could assure himself that members did not have any disagreements regarding provisional work plans.

Issues of Accessibility. Making oneself continually available on projects was a general problem experienced by all community members. Display practices, which facilitated the distributed engagement of diverse participants (at all times and from multiple locations), also expanded the breadth and depth of scrutiny and feedback that was feasible. This occasionally generated some tension in client interactions as these were mediated through the extranets. Use of extranets on projects increased the involvement of client members, but it also escalated a project's risk level because it increased dependency on client input. As information and ideas were made available to client members for their comments, Adweb members had to be willing and able to respond to these clients' ongoing feedback:

There are a lot more changes than in the past. We need to come to an understanding between us and them [the client]. At the beginning, a lot of clients don't understand how our websites are going to affect their workflow. Now that we are tying in with their systems, the website we design is very interactive with their business and has a big impact on how they do their work. But they often don't have a total picture of that. So there are a lot of changes once they get a more complete picture.

Moreover, it increased temporal dependence on clients, occasionally leading to delays:

It's a big problem when we don't get things from the client in time. Let's say we are doing design requirements. We'll say, "Here's what we think is a good solution. What do you think?" They need to ask their customers or ask people in their company if they have in-house development. If they don't turn it around quickly then it puts the project in jeopardy.

While Adweb members were accustomed to aggressive timelines, their clients were typically operating under different temporal pressures and could not always match those specified by the projects. Technology members, who had to engage extensively with clients' IT departments, were particularly vulnerable to delays depending on when and how clients interacted with them. As one Technology member put it:

Now that we are building technology with the client and are codependent with them...there is enormous dependency in the fabric we try to weave with them.

The practice of display as exercised through the use of online tools made more information available to all project members, but it also created constraints around expectations of continual accessibility and information processing. Members from all communities noted that there was pressure to be working "all the time" in order to avoid becoming the bottleneck on the fast-moving projects. For example, on the Insurance Inc. project, one Client Services member complained that she had repeatedly postponed a doctor's appointment for fear of missing her daily deadlines, and one of her colleagues on the project offered this illustration:

There's an expectation that you're available anytime, anywhere for anything.... People get out of control.... A person recently, for instance, talked to a [client] prospect and felt he really needed to redo a proposal and get it to them first thing the next morning. And he found that out at 7 P.M., and he could not understand why no one had responded to the e-mail that he'd sent out at 7 P.M. Now, I happened to be in at 5:30 the next morning. I redid the proposal and sent it to him by 7:30 A.M.,... but I had his boss down my throat at 9 A.M. that "You people are not being responsive."

Accessibility has also become an expectation of clients, as one Technology member observed:

The client can immediately call us on their cell, at any point and time, and get us. ... And in some ways it probably contributes to chaos because you hate to not answer the phone when the client's calling in. It raises our level of responsiveness probably a significant amount but the level of intrusiveness is affected.

In addition, Adweb members often had to adjust their temporal rhythms, particularly when clients were located

in different time zones. As a Client Services member noted, this meant longer hours for everyone:

It does mean that the hours have gotten crazy in the sense that we typically start conversations with them at 6 A.M. in the morning because the client is placed in London and it's working hours for them. And then they continue to call us until 9 or 10 P.M. in the night their time because it's working hours for us.

Compounding the pressure to be always accessible was the imperative to be constantly informed. Client Services and Project Management members often suggested that Creatives make more use of the repositories on the network. However, without adequate time or skills to do effective search and retrieval, Creative members perceived this activity as unproductive. As one member noted:

Sure, there is all this instant information, but how can you absorb it all? It's just like books. Initially there was the handset press and books were printed in a lower quantity. Now there are books by the million. I can buy more, they are less expensive, I can go to Amazon and get them right away. But am I reading more? No. You can drive a dump truck of information up to my house, but at the end of the day I am the weakest link. I have to sleep, I get colds, I need to cut the grass. If you give me 18 trillion pages of something, I can get through the same quantity as if you gave me 10,000. No one gives you a method for distilling this information. No one gives you the most salient bits. They just dump a phone book's worth on your desk.

The use of project e-mail augmented the capability of members to keep each other informed about what they were working on. Client Services and Project Management members used project e-mail extensively because it facilitated their communication with the team and the client, but the resulting volume of e-mail led to members receiving more communication than they could reasonably handle, often leaving them feeling overwhelmed, frustrated, and cynical. Two Technology members noted:

The buzzword is that we have to over-communicate: Give people information even if it isn't relevant—just in case.

People are over-informed, just cc'ed on everything. You get to a point that your potential for being informed gets [so] that you can't even read another [thing]. If it doesn't already have urgent-slash-crisis in the subject line, it becomes, at best, an e-mail that cries wolf type of thing.

These difficulties in the day-to-day management of project e-mail often led, paradoxically, to a lack of coordination by increasing ambiguity and undermining reflexivity. Because everyone on a project was copied on all messages, Project Management and Client Service members often assumed that everyone had been informed and understood what was being done. This assumption was often violated when other members hadn't received, read, or comprehended the messages, as this member noted:

As e-mail becomes the norm, you just become so reliant on communications being done that way. I think it ends up ultimately hurting relationships. I have huge issues with e-mail communication because you can't tell the tone, things get misread or they get glanced over, they don't really get agreed to, you don't necessarily get a response. You send out an e-mail and you assume everyone reads it, but if you're one of those people that gets 200 e-mails a day, you're not reading every single one.

Information overload also occurred in the detailed specificity with which some projects were planned and managed. Project Management members were responsible for devising detailed plans, scheduling numerous status meetings, and emphasizing project process and deadlines. This level of planning led some Creative members to complain that the result was a neglect of content in the interests of speed. As one Creative member noted:

I think [the level of planning] is overboard. ... We spend more time worrying about meeting dates than we do about content. We spend more time orchestrating than allowing the orchestra to play. If we took two projects and ran one without this detailed project management, it wouldn't cost anymore, it would be delivered on the same date, it would be a better product, and there would be better morale.

Not surprisingly, detailed plans and deadlines were sometimes resisted by members who felt their work could not be specified with such precision. A Technology member explained:

The project plans are a lot more detailed than they used to be. Now every item has a timeline. It helps Project Managers predict whether we can deliver on time on budget. But in reality the project plan changes every day. Because it is so detailed, there is not much buffer. People ignore the project plan. We just know the days when the big items are due. ... In practice, we don't follow these things in order. But it makes the Project Managers feel better.

Thus, when feeling overwhelmed with information or details, some members simply chose not to comply with deadlines or requirements set by Client Services and Project Management members. Most of the time, these omissions and workarounds were tolerated in the pragmatic interests of keeping the project moving forward. Because issues on projects were often short-lived and in flux, deviations typically did not have significant and far-reaching consequences. It was only when major difficulties were encountered (e.g., the Insurance Brazil breakdown) that attempts were made to comply more with detailed plans.

Issues of identity, control, and accessibility arose recurrently on the projects, and were reflected in the various ways in which members' everyday activities ignored, bypassed, modified, or refused to comply with expectations of cross-community coordination. While

such activities undermined cross-boundary coordination practices, they rarely disrupted the projects sufficiently to completely halt progress or produce irrecoverable failures. The urgency to keep moving and keep innovating helped to keep the action going on projects, thus reducing possibilities of members getting mired in jurisdictional contests. At the end of the day, what mattered for all these members was to get the work done and move on to the next project.

Discussion

Recently, scholars have suggested that existing organizational paradigms, based as they are on assumptions of stability and inertia, may be less useful in helping us make sense of emerging organizational phenomena (Barley and Kunda 2001, Daft and Lewin 1993, Weick and Quinn 1999). Child and McGrath (2001, p. 1135), for example, argue "there is a widespread gap between the rapid development of new organizational forms in practice and the capacity of existing perspectives to account for them in theory." In our field study of one heterarchic organization, we found that existing perspectives on knowledge sharing and coordination do not adequately explain the cross-boundary coordination practices that we observed in the everyday work activities of members engaged in fast-paced and volatile Web-based projects.

Enacting a Trading Zone

The literature on knowledge sharing suggests that organization members from diverse specialties can best work across community boundaries when they: (i) share a common lexicon; (ii) help to reconcile interpretive differences by creating shared meaning; and (iii) facilitate means through which individuals can jointly transform their local knowledge. Carlile (2002) refers to these three processes as *transferring*, *translating*, and *transforming*. This is a valuable framework, but our findings suggest that it needs augmenting to account for a different set of activities and accommodations that we found accomplished cross-boundary coordination in horizontal, fast-paced, and heterogeneous conditions. Thus, rather than transferring local information, organizational actors make their work and commitments visible to others through posting to a common digital space that others can access. Instead of translating local knowledge through linguistic conversion and cultural adaptation, organizational actors render their heterogeneous ideas in a form that is legible (if not understandable) to others. Also, instead of transforming local understandings into shared meanings and common knowledge, organizational actors juxtapose their diverse efforts into a provisional and emerging collage of loosely coupled contributions. We identified these performances as practices of display, representation, and assembly.

To develop these ideas further, we draw on a perspective developed by Galison (1997, 1999) to make sense of the complex work of physicists. In explaining how distinct communities within physics—in particular, theorists, experimentalists, and engineers—aligned their activities without homogenizing the inherent diversity of their community interpretations, identities, and interests, Galison finds that despite fundamental differences, members of these communities were able to work out “exchanges,” or what we would call coordination practices, “in exquisite local detail, without global agreement” (Galison 1997, p. 46). He proposes an understanding of such cross-disciplinary interaction by developing the metaphor of a *trading zone* to highlight how the local coordination of ideas and actions may take place despite differences in community purposes, norms, meanings, values, and performance criteria. Importantly, this metaphor is not intended to evoke the commodified transactions of efficient markets, but the complex interactions of distinct communities encountering each other for purposes of exchange. As Galison notes (1997, p. 803):

I intend the term “trading zone” to be taken seriously, as a social, material, and intellectual mortar binding together the disunified traditions of experimenting, theorizing, and instrument building. Anthropologists are familiar with different cultures encountering one another through trade, even when the significance of the objects traded—and of the trade itself—may be utterly different for the two sides.

In Galison’s analysis, enacting a trading zone does not require equivalence or similarity of interpretations or interests, nor does it assume stability or permanence of relations. Instead, members of different communities coordinate their actions temporarily and locally, navigating their differences in norms, meanings, and interests only as needed. Engaging in a trading zone suggests that diverse groups can interact across boundaries by agreeing on the general procedures of exchange even while they may have different local interpretations of the objects being exchanged, and may even disagree on the intent and meaning of the exchange itself. Such an understanding evokes a view of cross-boundary coordination as performative, as emergent in recurrent actions, and thus as a provisional and ongoing accomplishment. Vaughan (1999), who has used Galison’s notion of a trading zone to study NASA engineers and their subcontractors, argues that such interactions are particularly useful in conditions of uncertainty and change because they do not depend on shared ideas, interests, or norms, which are difficult to accomplish when time is short, meanings are divergent, and conditions are ambiguous.

We believe the metaphor of a trading zone is a particularly useful lens with which to understand the cross-boundary coordination that occurred regularly on Adweb projects. Building on Galison (1997) and Vaughan

(1999), we propose that the practices of display, representation, and assembly engaged in by Adweb members can be seen to enact a trading zone that facilitated their temporary, local, ongoing, and dynamic cross-boundary coordination. We propose understanding this trading zone as a *coordination structure* that facilitates cross-boundary coordination in fast-paced, temporary, and volatile conditions. Such a coordination structure is recurrently enacted through the ongoing activities of organizational actors as they engage in cross-boundary practices to make their work visible, legible, and aligned. Structure in this view is understood as an ongoing accomplishment, emerging from actors’ continuous engagement in everyday life, rather than a static property of social systems (Giddens 1984).

The enactment of a trading zone is an ongoing accomplishment that depends on members’ temporal accommodations and resistances (Pickering 1995) as they engage with each other and their technologies. Because the trading zone is “always in the making,” cross-boundary coordination is a contingent, emergent, and dynamic outcome that cannot be planned or prescribed, but is highly dependent on the situated activities of the various communities. We believe that this view of cross-boundary coordination as the ongoing enactment of a trading zone is useful in understanding the fast-paced, temporary, and decentralized workplaces that are characteristic of many contemporary forms of organizing (see Table 4).

The trading zone enacted through practices of display, representation, and assembly entailed a set of norms, interpretations, and facilities that Adweb members produced/reproduced in their everyday project activities. Even though Adweb members were colocated in the same building, and in many cases in the same hallway, the trading zone they enacted on their projects was extensively mediated by digital technologies. Adweb’s distinctive use of such technologies (what Orlikowski 2000 calls technology-in-practice) to accomplish their cross-boundary coordination practices differed from the face-to-face medium and various tangible artifacts used by Galison’s physicists to enact their scientific trading zone. It also departed from the practice of technology brokering described by Hargadon and Sutton (1997), where individual designers served as the carriers of ideas over time, keeping their own collections of interesting technologies (e.g., Velcro and high-strength alloys) to use on later projects. At Adweb, in contrast, members made their past and present materials visible to all on the intranet, and any Adweb member could access these and merge them in new ways to reuse on current projects. As these materials are digitally rendered, they can be “reshuffled and recombined” in various ways across time and space (Latour 1988, p. 45), reflecting and reproducing Adweb’s networked and disembodied mode of organizing.

Table 4 Enacting a Trading Zone Within Adweb

Practices	Activities	Technologies	Consequences	Issues
Display	Rendering work visible to others on the project Making schedules and plans available to others	Project E-mail (Outlook) Intranet, Extranets Calendar (Outlook) Project Management (Project)	Keeping information and ideas transparent Facilitating speed and distributed authority Supporting ongoing revision and alignment	Identity Control Accessibility
Representation	Rendering work legible through use of project genres, e.g., <i>PowerPoint Presentation, Discovery Matrix, Discover Document, Proof of Concept</i>	Project E-mail (Outlook) Intranet, Extranets PowerPoint Word, Excel	Facilitating sharing of ideas and information through making it legible to others Embedding flow of work and responsibilities in genres	Identity Control
Assembly	Juxtaposing existing work through modification and recomposition Reusing prior work Aligning through provisional settlements	Project E-mail (Outlook) Intranet, Extranets PowerPoint Word, Excel	Creating collage of diverse contributions Leveraging prior knowledge Keeping work adaptable through fluid agreements	Identity Control Accessibility

Adweb members' use of information technologies was clearly constitutive of the cross-boundary coordination structure they enacted on their projects, and it both enabled important capabilities, but also constrained certain activities, reflections, and outcomes. Adweb members' use of electronic tools shaped their cross-boundary coordination in important ways, distinguishing it from project work with much less dependence on information technology (e.g., construction projects, movie sets). Without online networks, application tools, and pervasive e-mails, Adweb members would not have been able to keep each other informed through constant online display of their work. They would not have been able to access, reuse, revise, and assemble each other's work to the same extent and with the same speed and facility, and they would not have engaged in detailed ongoing interactions with client members. The use of technologies to enact Adweb's trading zone, thus, afforded rapid response to changing requirements, technologies, and inputs; flexibility in designs and decisions; distributed forms of authority and accountability; and increased participation from diverse players. Such technology use, however, also constrained the kind of trading zone that was enacted, and shaped its possible consequences.

The continual revisions to ideas and information made possible through round-the-clock access to the intranet and extranets increased Adweb members' feelings of overload and excessive accessibility. It also increased the risk of inattention and incomprehension as members were inundated with information. As Perlow (1999) found in her study of engineers, frequent interruptions can create a climate of crisis, and perpetuate feelings of temporal pressure. Adweb's norms of constant availability similarly perpetuated a sense of unrelenting temporal pressure for Adweb members, which led to difficulties in how members coped with competing demands on

their time and in their ability to distinguish urgent or important information from the rest, particularly when all e-mail messages took the same form. The flattening of content in e-mails and presentations, while facilitating projectwide legibility, inhibited the texture and richness that is often the stuff of creative friction, thus potentially stifling innovation. The extent of visibility on a given project reflected the power balance, which was constantly shifting between projects and even within a given project over time (depending on which community was taking the lead at any time). While effective display of work progress and schedules often saved time and effort, decisions about visibility were typically shaped in contests among communities battling over opportunities to enhance status, assert identity, leverage expertise, and protect interests.

Engaging with Boundary-Spanning Mechanisms

In their enactment of their project's trading zone, Adweb members encountered each other through their collective interaction with each other's electronic files, messages, and code. Such virtual encounters resemble the disembodied performances identified by Child and McGrath (2001) as more evident in contemporary postbureaucratic organizations. This loose engagement with electronic representations, along with expectations of legibility, transparency, and distributed assembly that entailed provisional settlements and ongoing adjustments, allowed Adweb members to interact on projects without making deep commitments to shared meanings or transformed knowledge. While such a lack of integration sometimes created problems and challenges for the project teams, it also facilitated coordination in a context of rapid change, where client requirements and technologies were

ill-defined and dynamic, and where speed and adaptability were privileged over mutual understanding and shared commitment.

The privileging of form rather than content in Adweb's representation practices is consistent with earlier organizational research that suggests shared meanings are not essential for collective learning and organized action (Donnellon et al. 1986, Hutchins 1995, Star and Griesemer 1989, Weick 1979). For example, Fiol (1994, p. 403) found that venture development teams were able to generate "unified ways of framing their arguments, while at the same time maintaining diversity through differences in the content of team members' interpretations." Similarly, Donnellon et al. (1986, p. 44) discovered that a "shared repertoire of communication behaviors" enabled groups to engage in organized action in the absence of shared meaning. In their analyses, group members' use of shared communication mechanisms helped them to generate equifinal meanings, that is, "interpretations that are dissimilar but that have similar behavioral implications" (p. 44), and these were sufficient for effective collective action.

These findings are echoed in our observations that Adweb members' use of a repertoire of project genres enabled their coordination across boundaries. The use of common project genres let members of each of Adweb's four communities represent their ongoing work while reducing the nuances of their particular local concerns. This afforded a level of discursive uniformity that allowed project members to quickly focus on the salient issues, without requiring a thorough understanding of the various underlying details specific to particular communities. While this flattening of content repressed some community-specific interests and interpretive differences, it also helped Adweb members communicate key ideas across community boundaries, thus enabling ongoing interaction on the project. Such use of genres to coordinate work across time and space resembles Turnbull's (1993, 2000) findings that the use of physical templates was critical to the work of craftsmen who built the complex and radically innovative Gothic cathedrals, a feat achieved in those medieval times without a common system of measurement, without drawn plans, and without the expertise of architects. While these craftsmen were not dealing with rapidly changing conditions, they were faced with numerous design changes and considerable complexity and heterogeneity. Turnbull argues that the value of templates in such conditions lay, in part, in their minimal requirements for communication. In contrast to site drawings that would have demanded commitments to a standard system of measures, templates served as exemplars that could be applied to multiple problems and situations. Genres, similarly, are expressive templates that represent ways of communicating across communities to achieve collective purposes.

In existing studies of new product development within traditional organizations, members from different communities often use boundary objects to inscribe and share meanings across their different local contexts without requiring a fully joint definition of the object (Bechky 2003a, Carlile 2002, Henderson 1991, Star and Griesemer 1989). Boundary objects (e.g., structural drawings) are robust enough to maintain a common identity across sites, but flexible enough to be used for local purposes within different social worlds (e.g., engineering firms, architectural offices, construction sites). Carlile (2002) describes how boundary objects provide a concrete method for individuals to specify and learn about their differences and dependencies across specific boundaries, and are useful for navigating boundaries by enabling the representation and transformation of knowledge. Bechky (2003a, b) highlights that boundary objects enable knowledge sharing when used to create common understandings, but that they may constrain it when used to legitimize or impose occupational interests.

While genres might be used as boundary objects in some contexts, our work at Adweb suggests that in the fast-paced and dynamic world of Web-based product development, they were not. In Star and Griesemer's (1989) definition, boundary objects are weakly structured across communities, and strongly structured within communities. They are effectively used when interpretations, values, and interests are sufficiently stabilized so that "differences and dependencies" can be acknowledged and accepted (Carlile 2004). In such situations, instances of (at least partial) closure are possible because individuals can negotiate with one another as they take into account each other's viewpoints when reaching a decision about what is to be represented in the boundary objects (Star and Griesemer 1989). The result, often, is a strongly structured object that can be extended across a wide array of contexts (Henderson 1991).

In contexts such as Adweb, change is not iterative, but continuous. Thus, attempts to construct shared mechanisms such as boundary objects are difficult because ideas and knowledge are changing too rapidly to be effectively inscribed within objects. We did not find that Adweb members strongly structured their representations, either within or across their community boundaries. Indeed, doing so would have undermined their imperatives to be fast, flexible, and responsive. As Sapsed and Salter (2004, p. 1531) found in their study of a global, dispersed computing project, boundary objects in "high-pressure, project-based" contexts are "...more limited in their effects," in part because they are "high-maintenance" and insufficiently flexible for the unpredictable demands of volatile conditions. Similarly, Adweb's high-pressure projects were conducted in conditions that do not easily generate the common ground that helps transform local understandings to achieve effective knowledge sharing (Bechky 2003a, Carlile 2002).

Some of Adweb's cross-boundary interactions were conducted across firm boundaries as members engaged with clients throughout their projects. The relative permeability of the boundaries between Adweb and its clients reflects both the heterarchic nature of Adweb's form of organizing and its status as a service organization. While we were unable to systematically analyze client engagement with Adweb, it was evident that such engagement critically shaped members' ongoing coordination practices. Examining the nature, role, and consequences of such client coproduction represents an important area of future research.

There are important differences between Adweb projects and those studied in the existing literature that can help explain the differences in findings. Prior work on cross-boundary coordination has tended to involve manufacturing or professional work, while Adweb's projects were located in the new arena of website development. Prior work describes projects that exhibited well-established hierarchies and strong occupational distinctions, while Adweb projects were more loosely coupled, reflected greater heterogeneity, and were situated within a rapidly changing context. Furthermore, compared to projects studied by other scholars, Adweb's projects were more temporary (on the order of weeks rather than months or years), smaller in size (involving only a handful of members), more unsettled (conditions were often unpredictable), and less tangible (depending on digital media rather than physical artifacts).

Given these important differences, it is not surprising that our exploratory study found coordination within Adweb different than that articulated in the existing literature. In particular, we argue that coordination across boundaries in conditions such as those within Adweb may be more usefully understood as being accomplished through the ongoing enactment of a trading zone where community members make information visible rather than transferring it from one community to another, where they represent their work legibly through a repertoire of project genres rather than translating different meanings to reconcile interpretive differences, and where they assemble work products through a process of juxtaposition, adaptation, and dynamic alignment, rather than engaging in joint transformation.

Our study was exploratory and limited to a single firm. Furthermore, the particular firm we studied had very distinctive characteristics, and we studied it during a volatile time in its industry (the dotcom boom and bust). These factors necessarily qualify our insights. However, to the extent that this context reflects those we might expect in the future as more firms attempt to operate in more dynamic and technology-mediated ways, our exploratory findings may offer some useful initial insights. Clearly, future research is required to further examine the initial relationships and patterns outlined here.

Conclusion

Existing perspectives on cross-boundary coordination have characterized the difficulties associated with sharing knowledge across boundaries, and the activities and artifacts that communities use to do so. Our empirical study of an interactive marketing organization contributes to this literature by highlighting how boundary spanning in rapidly changing and digitally mediated environments is accomplished through actors' enactment of trading zones in their ongoing, technology-based coordination practices. As individuals draw on diverse emerging interests and information to accomplish their complex, dynamic, and heterogeneous project-based work, they use technology to coordinate their actions and ideas across community boundaries. This enables them to develop new Web-based products by making work and commitments visible, rendering their work legible by (re)producing a repertoire of project genres, and assembling their work through juxtaposition, reuse, and ongoing alignment. While these practices may facilitate adaptability, speed, and learning in the short term, they also entail contradictions associated with community members' attempts to hold onto their local knowledge, social identities, and perceived interests as they work across boundaries. These contradictions may limit organizational creativity in the long run by suppressing the mutual engagement and occupational friction that can generate creativity. Because these contradictions are inextricably linked to the ongoing practices of working across boundaries, they are deeply implicated in the new forms of organizing, and constitute part of the texture of emerging modes of boundary-spanning, technology-based work. This has important implications for how we understand postbureaucratic organizing in general, and cross-boundary coordination in particular.

Acknowledgments

The authors thank the members of Adweb who participated in this study, as well as the senior editor and reviewers for their valuable guidance during the revision process. They are also grateful for the helpful comments they received on earlier versions of this manuscript from Lotte Bailyn, Pablo Boczkowski, and Martha Feldman. This research was supported by a grant from the National Science Foundation under Award IIS-0085725.

Endnotes

¹Names of the company, its members, clients, and proprietary methods have been disguised for confidentiality reasons.

²After our study in 2001, Adweb laid off 25% of its workforce and subsequently merged with a similar company in 2003. It still operates under the Adweb masthead, with approximately 500 employees in five global offices.

³Space constraints have led us to focus primarily on crossing internal community (rather than external firm) boundaries in this paper.

References

- Adler, P. S. 1995. Interdepartmental interdependence and coordination: The case of the design/manufacturing interface. *Organ. Sci.* 6(2) 147–167.
- Agar, M. 1980. *The Professional Stranger: An Informal Introduction to Ethnography*. Academic Press, New York.
- Allen, T. 1971. Communication, technology transfer, and the role of the technical gatekeeper. *R&D Management* 1 14–21.
- Barley, S. R. 1996. Technicians in the workplace: Ethnographic evidence for bringing work into organization studies. *Admin. Sci. Quart.* 41(3) 404–441.
- Barley, S. R., G. Kunda. 2001. Bringing work back in. *Organ. Sci.* 12(1) 76–95.
- Bechky, B. A. 2003a. Object lessons: Workplace artifacts as representations of occupational jurisdiction. *Amer. J. Sociology* 109(3) 720–752.
- Bechky, B. A. 2003b. Sharing meaning across occupational communities: The transformation of understanding on a production floor. *Organ. Sci.* 14(3) 312–330.
- Boland, R. J., Jr., R. V. Tenkasi. 1995. Perspective making and perspective taking in communities of knowing. *Organ. Sci.* 6 350–372.
- Borer, M. L. 2002. The Cyborgian self: Toward a critical social theory of cyberspace. *Reconstruction: Stud. Contemporary Culture* 2(3). <http://www.reconstruction.ws/023/borer.htm>.
- Brown, J. S., P. Duguid. 1991. Organizational learning and communities-of-practice: Toward a unified view of working, learning, and innovation. *Organ. Sci.* 2(1) 40–57.
- Brown, J. S., P. Duguid. 1998. Organizing knowledge. *California Management Rev.* 40(3) 90–111.
- Callon, M., ed. 1998. *The Laws of the Markets*. Blackwell Publishers, Oxford, UK.
- Callon, M. 2002. Writing and (re)writing devices as tools for managing complexity. J. Law, A. Mol, eds. *Complexities: Social Studies of Knowledge Practices*. Duke University Press, Durham, NC, 191–217.
- Carlile, P. R. 2002. A pragmatic view of knowledge and boundaries: Boundary objects in new product development. *Organ. Sci.* 13(4) 442–455.
- Carlile, P. R. 2004. Transferring, translating, and transforming: An integrative framework for managing knowledge across boundaries. *Organ. Sci.* 15(5) 555–568.
- Castells, M. 1996. *The Rise of the Network Society*. Blackwell Publishers, Malden, MA.
- Child, J., R. G. McGrath. 2001. Organizations unfettered: Organizational form in an information-intensive economy. *Acad. Management J.* 44(6) 1135–1148.
- Ciborra, C. U. 1996. The platform organization: Recombining strategies, structures, and surprises. *Organ. Sci.* 7(2) 103–118.
- Daft, R. L., A. Y. Lewin. 1993. Where are the theories for the “new” organizational forms? *Organ. Sci.* 4(6) i–v.
- Donnellon, A., B. Gray, M. G. Bougon. 1986. Communication, meaning, and organized action. *Admin. Sci. Quart.* 31(1) 43–55.
- Dougherty, D. 1992. Interpretive barriers to successful product innovation in large firms. *Organ. Sci.* 3(2) 179–202.
- Eisenhardt, K. M. 1989. Building theories from case study research. *Acad. Management Rev.* 14(4) 532–550.
- Fiol, C. M. 1994. Consensus, diversity, and learning in organizations. *Organ. Sci.* 5(3) 403–420.
- Galbraith, J. R. 1973. *Designing Complex Organization*. Addison-Wesley Publishing, Reading, MA.
- Galison, P. 1997. *Image and Logic: A Material Culture of Microphysics*. University of Chicago Press, Chicago, IL.
- Galison, P. 1999. Trading zone: Coordinating action and belief. M. Biagioli, ed. *The Science Studies Reader*. Routledge, New York, 137–160.
- Gherardi, S., D. Nicolini. 2000. To transfer is to transform: The circulation of safety knowledge. *Organization* 7(2) 329–348.
- Giddens, A. 1984. *The Constitution of Society*. Polity Press, Cambridge, UK.
- Giddens, A. 1991. *Modernity and Self-Identity: Self and Society in the Late Modern Age*. Stanford University Press, Stanford, CA.
- Girard, M., D. Stark. 2002. Distributing intelligence and organizing diversity in new-media projects. *Environ. Planning A* 34(11) 1927–1949.
- Glaser, B. G., A. L. Strauss. 1967. *The Discovery of Grounded Theory: Strategies for Qualitative Research*. Aldine Publishing, Chicago, IL.
- Grabher, G. 2001. Ecologies of creativity: The village, the group, and the heterarchic organisation of the British advertising industry. *Environ. Planning A* 33(2) 351–374.
- Grabher, G. 2002. Fragile sector, robust practice: Project ecologies in new media. *Environ. Planning A* 34(11) 1911–1926.
- Grant, R. M. 1996. Prospering in dynamically competitive environments: Organizational capability as knowledge integration. *Organ. Sci.* 7(4) 375–387.
- Hansen, M. T. 1999. The search-transfer problem: The role of weak ties in sharing knowledge across organization subunits. *Admin. Sci. Quart.* 44(1) 82–111.
- Hargadon, A., R. Sutton. 1997. Technology brokering and innovation in a product development firm. *Admin. Sci. Quart.* 42(4) 716–749.
- Heckscher, C. C., A. Donnellon, eds. 1994. *The Post-Bureaucratic Organization: New Perspectives on Organizational Change*. Sage Publications, Thousand Oaks, CA, 129–177.
- Henderson, K. 1991. Flexible sketches and inflexible data-bases: Visual communication, inscription devices, and boundary objects in design engineering. *Sci. Tech. Human Values* 16(4) 448–473.
- Heydebrand, W., A. Miron. 2002. Constructing innovativeness in new-media start-up firms. *Environ. Planning A* 34(11) 1951–1984.
- Hutchins, E. 1995. *Cognition in the Wild*. MIT Press, Cambridge, MA.
- Iskander, N. 2005. Innovating state practices: Migration, development, and state learning in the Moroccan Souss. Working paper, Industrial Performance Center, Massachusetts Institute of Technology, Cambridge, MA.
- Kotamraju, N. 1999. The birth of web site design skills: Making the present history. *Amer. Behavioral Scientist* 43(3) 464–474.
- Lant, T. K. 1999. The creation of cognitive communities in an emerging industry: The New York new media industry. *Proc. Soc. Organ. Learning Res. Greenhouse*. Cambridge, MA.
- Latour, B. 1988. Drawing things together. M. Lynch, S. Woolgar, eds. *Representation in Scientific Practice*. MIT Press, Cambridge, MA, 19–68.
- Lave, J. 1988. *Cognition in Practice*. Cambridge University Press, Cambridge, UK.
- Lave, J., E. Wenger. 1991. *Situated Learning: Legitimate Peripheral Participation*. Cambridge University Press, Cambridge, UK.

- Lawrence, P. R., J. W. Lorsch. 1969. *Organization and Environment: Managing Differentiation and Integration*. Richard D. Irwin, Homewood, IL.
- Leonard-Barton, D. 1992. Core capabilities and core rigidities: A paradox in managing new product development. *Strategic Management J.* **13** 111–125.
- Levina, N. 2001. Multi-party information systems development: The challenge of cross-boundary communication. Unpublished doctoral dissertation, Massachusetts Institute of Technology, Cambridge, MA.
- Neff, G., D. Stark. 2003. Permanently Beta: Responsive organization in the Internet era. P. Howard, S. Jones, eds. *Society Online: The Internet in Context*. Sage Publications, Thousand Oaks, CA, 173–188.
- Nelson, R., S. Winter. 1982. *An Evolutionary Theory of Economic Change*. Harvard University Press, Cambridge, MA.
- Nonaka, I. 1994. A dynamic theory of organizational knowledge creation. *Organ. Sci.* **5**(1) 14–38.
- Orlikowski, W. J. 2000. Using technology and constituting structures: A practice lens for studying technology in organizations. *Organ. Sci.* **11**(4) 404–428.
- Orlikowski, W. J. 2002. Knowing in practice: Enacting a collective capability in distributed organizing. *Organ. Sci.* **13**(3) 249–273.
- Orlikowski, W. J., J. Yates. 1994. Genre repertoire: Examining the structuring of communicative practices in organizations. *Admin. Sci. Quart.* **39**(4) 541–574.
- Orlikowski, W. J., J. Yates, K. Okamura, M. Fujimoto. 1995. Shaping electronic communication: The metastructuring of technology in the context of use. *Organ. Sci.* **6**(4) 423–444.
- Orr, J. E. 1996. *Talking about Machines: An Ethnography of a Modern Job*. ILR, Ithaca, NY.
- Pawlowski, S., D. Robey. 2004. Bridging user organizations: Knowledge brokering as the work of information technology professionals. *MIS Quart.* **28**(4) 645–672.
- Perlow, L. 1999. The time famine: Towards a sociology of work time. *Admin. Sci. Quart.* **44** 57–81.
- Pickering, A. 1995. *The Mangle of Practice: Time, Agency and Science*. The University of Chicago Press, Chicago, IL.
- Powell, W. 1996. Interorganizational collaboration in the biotechnology industry. *J. Institutional Theoret. Econom.* **120** 197–215.
- Powell, W. 2001. The capitalist firm in the 21st century: Emerging patterns. P. DiMaggio, ed. *The Twenty-First Century Firm: Changing Economic Organization in International Perspective*. Princeton University Press, Princeton, NJ.
- Rindova, V. P., S. Kotha. 2001. Continuous “Morphing:” Competing through dynamic capabilities, form, and function. *Acad. Management J.* **44**(6) 1263–1280.
- Sapsed, J., A. Salter. 2004. Postcards from the edge: Local communities, global programs, and boundary objects. *Organ. Stud.* **25**(9) 1515–1534.
- Schatzki, T. R., K. Knorr-Cetina, E. Savigny. 2001. *The Practice Turn in Contemporary Theory*. Routledge, London, New York.
- Schon, D. A. 1983. *The Reflective Practitioner: How Professionals Think in Action*. Basic Books, New York.
- Simon, H. A. 1982. *Models of Bounded Rationality*. MIT Press, Cambridge, MA.
- Sitkin, S. B., J. S. Brown. 1999. Fostering the coupling of generative differences: Creating space for pluralism. *Proc. AOM 1999 Conf.*, Chicago, IL.
- Spender, J. C. 1996. Making knowledge the basis of a dynamic theory of the firm. *Strategic Management J.* **17** 45–62.
- Star, S. L., J. R. Griesemer. 1989. Institutional ecology, translations, and boundary objects: Amateurs and professionals in Berkeley’s Museum of Vertebrate Zoology, 1907–1939. *Soc. Stud. Sci.* **19** 387–420.
- Stark, D. 1999. Heterarchy: Distributed authority and organizing diversity. J. H. Clippinger, ed. *The Biology of Business: Decoding the Natural Laws of Enterprise*. Jossey-Bass, San Francisco, CA, 153–179.
- Stark, D. 2001. Ambiguous assets for uncertain environments: Heterarchy in postsocialist firms. P. DiMaggio, ed. *The Twenty-First Century Firm: Changing Economic Organization in International Perspective*. Princeton University Press, Princeton, NJ.
- Strauss, A. L., J. M. Corbin. 1990. *Basics of Qualitative Research: Grounded Theory Procedures and Techniques*. Sage Publications, Newbury Park, CA.
- Suchman, L. 1994. Working relations of technology production and use. *CSCW: J. Collaborative Comput.* **2** 21–39.
- Teece, D. J. 1998. Capturing value from knowledge assets: The new economy, markets for know-how, and intangible assets. *California Management Rev.* **40**(3) 55–79.
- Tsoukas, H. 1996. The firm as a distributed knowledge system: A constructionist approach. *Strategic Management J.* **17** 11–25.
- Turnbull, D. 1993. The ad hoc collective work of building Gothic cathedrals with templates, string, and geometry. *Sci., Tech., Human Values* **18**(3) 315–340.
- Turnbull, D. 2000. *Masons, Tricksters and Cartographers: Comparative Studies in the Sociology of Scientific and Indigenous Knowledge*. Routledge, London, UK.
- Vaughan, D. 1999. The role of the organization in the production of techno-scientific knowledge. *Soc. Stud. Sci.* **29**(6) 913–943.
- Volberda, H. W. 1996. Toward the flexible form: How to remain vital in hypercompetitive environments. *Organ. Sci.* **7**(4) 359–374.
- Weick, K. E. 1979. *The Social Psychology of Organizing*. McGraw Hill, New York.
- Weick, K. E., R. E. Quinn. 1999. Organizational change and development. *Annual Rev. Psych.* **50** 361–386.
- Wenger, E. 1998. *Communities of Practice: Learning, Meaning, and Identity*. Cambridge University Press, Cambridge, UK.
- Wittel, A., C. Lury, S. Lash. 2002. Real and virtual connectivity: New media in London. S. Woolgar, ed. *Virtual Society? Technology, Cyperbole, Reality*. Oxford University Press, Oxford, UK, 189–208.
- Yanow, D. 2000. Seeing organizational learning: A “cultural” view. *Organization* **7**(2) 247–268.
- Yates, J., W. J. Orlikowski. The PowerPoint presentation and its corollaries: How genres shape communicative action in organizations. M. Zachry, C. Thralls, eds. *Communicative Practices in Workplaces and the Professions: Cultural Perspectives on the Regulation of Discourse and Organizations*. Baywood Publishing, Amityville, NY. Forthcoming.
- Yates, J., W. J. Orlikowski, K. Okamura. 1999. Explicit and implicit structuring of genres: Electronic communication in a Japanese R&D organization. *Organ. Sci.* **10**(1) 83–103.

Copyright 2006, by INFORMS, all rights reserved. Copyright of Organization Science: A Journal of the Institute of Management Sciences is the property of INFORMS: Institute for Operations Research and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.